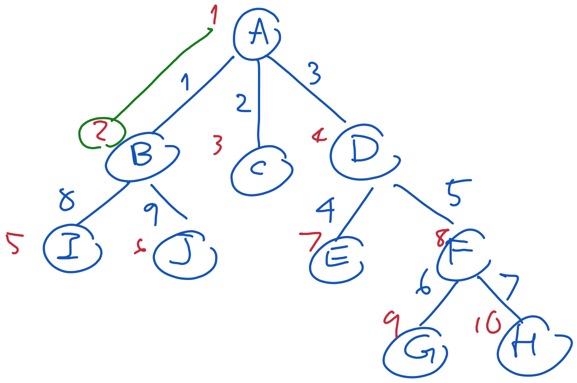
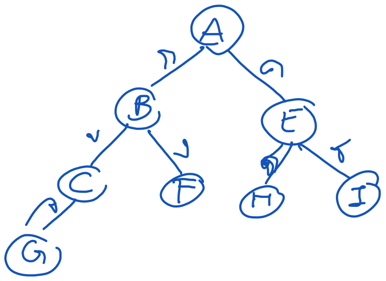
Decision Tree

Friday, October 16, 2026 11:38 AM



id	Parent (Int)	Node	Bran
1	0	A	–
2	1	B	1
3	1	C	2
4	1	D	3
5	2	I	8
6	2	J	9
7	4	E	4
8	4	F	5
9	8	G	6
10	8	H	7
		✓	✓

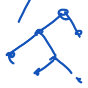


	Parent	Node	Bran
1	0	A	–
2	1	B	ก
3	2	C	ข
4	1	E	ค
5	2	F	ง
6	3	G	จ
7	4	H	ฉ
8	4	I	ช ✓

Outlook

CART

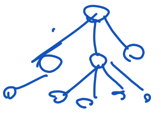
① binary tree



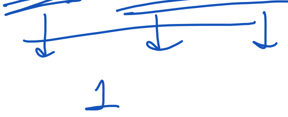
② gini/IG → v×Attr

③ reuse A ✓

ID3



{Sunny, Overcast, rain}



IG = Attr

reuse A X จนสิ้นเส้นทาง

Exam

2 Hr.

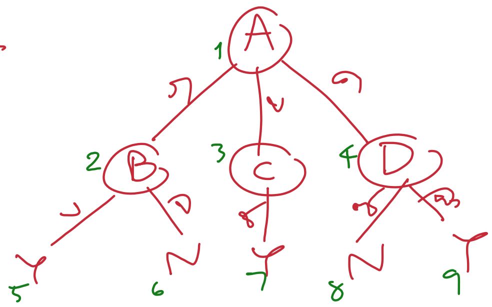
- A4 1 แผ่น (หน้า-หลัง) ลายมือ
- Calculator (No Programming)
- Pencil/Pen

Regression
- linear
- Polynomial
- k-NN

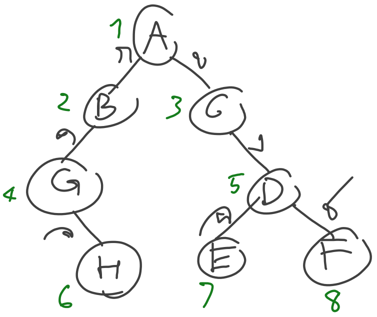
Classification
- k-NN → Euclidean, Manhattan
- perceptron (Vector)
- FFBP / Matrix form
- ID3 / Data Structure

Clustering
- k-Means

PM



id	Parent	Node	Branch
1	0	A	—
2	1	B	ก
3	1	C	ข
4	1	D	ค
5	2	Y	ง
6	2	N	จ
7	3	Y	ช
8	4	N	ซ
9	4	Y	ฉ



#	parent	Node	Branch
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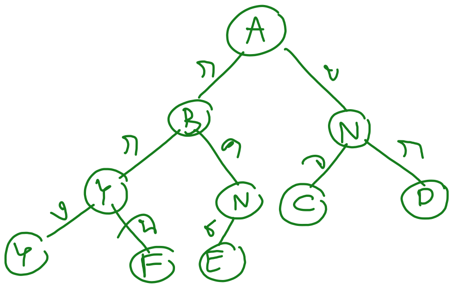
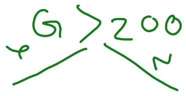
Parent	Node	Branch
0	A	—
1	B	ก
2	Y	ก
1	N	ข
2	N	ค
3	Y	ง
4	C	จ
4	D	ก
5	E	ช
3	F	ฉ



CART

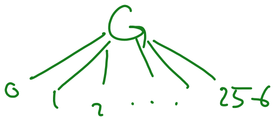
- Binary tree ←
- Node = Question
  Y / N



- Criteria = Gini $1-p^2$ ~~Gain~~
- Continue.

ID3 → C4.5, C

- n-ary tree (#Vattr)
- Node = Attr name



- Gain
- Descrete

- Regression
- Classification